

OPEN-SOURCE PIPELINE · HINDI TECHNICAL LECTURES

Technical lectures in Hindi.

VidVaani converts English technical lectures into timing-aligned Hindi audio while preserving technical vocabulary and the original video.

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Motivation

50,000+

hours of NPTEL engineering lectures

1 command

from a YouTube link or local video to a Hindi dub

₹7–11

measured cost for a seven-minute lecture

For many students, spoken English limits how effectively a technical lecture can be reused. The aim is a **low-cost Hindi version** that remains faithful to the teacher's terminology and timing.

Limitations of existing options

PLATFORM

YouTube

YouTube determines channel eligibility. Nipun's lecture channel is not enabled. Dubs cannot be edited, and instructors cannot choose terminology or voice.

SAAS

Commercial services

Typical prices are ₹3,000–11,000 per lecture, often with monthly limits and additional charges for each language.

HUMAN WORKFLOW

NPTEL / Bhashini

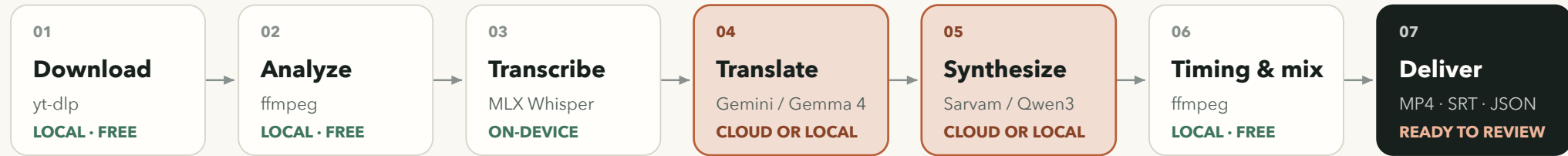
These programmes provide careful review, but operate course by course and are not self-service tools for instructors.

Need: a self-service system with explicit control over voice, vocabulary, timing, and cost.

System overview

ONE INSPECTABLE RUN

URL or local file → MP4 · SRT · JSON



Only these two phases need an API—and both have working local alternatives.

Local STT

MLX Whisper performs transcription on-device

Cached

a new voice can reuse earlier results

Measured

time, tokens, cost and output metadata are recorded

Inspecting a completed run

VIDVAANI

PROCESS CONTROL

Start processing

→

Explore completed sample

NO API CALLS

Recent runs

0

Completed runs will remain here during this server session.

LOCAL PIPELINE

Runtime ready

06

Timing & mix

ffmpeg

Timing & mix complete

25s

✓

07

Deliverables

MP4 · SRT · JSON

Ready – sample video available

0.1s

✓

PHASE 04 · COMPLETE

Follow live

Hindi translation

Translate each timed English segment into spoken Hindi while protecting technical vocabulary and the time budget.

PHASE PROGRESS

100%

STATE

Complete

TIME TAKEN

00:17

SEGMENTS

5 / 5

MODEL

gemini-2.5-flash

INPUT TOKENS

815

OUTPUT TOKENS

818

TRANSLATION COST

\$0.002289

Aligned translation

3 segments shown

00:00–00:15

ENGLISH

Today I want to show the usage of a tool called GeoGebra.

HINDI

आज मैं जिओजेब्रा नामक एक टूल का उपयोग दिखाना चाहता हूँ।

00:15–00:28

ENGLISH

We are given this uniform joint PDF in two dimensions.

HINDI

हमें दो आयामों में यह यूनिफॉर्म जॉइंट प्रोबैबिलिटी फंक्शन दिया गया है।

00:28–00:40

ENGLISH

The x range is between zero to two and the y range is between zero to two.

HINDI

x और y की रेंज शून्य से दो तक है।

5

Translation for a technical classroom

00:00–00:15

ENGLISH

Today I want to show the usage of a tool called GeoGebra.

HINDI

आज मैं जियोजेब्रा नामक एक टूल का उपयोग दिखाना चाहता हूँ।

00:28–00:40

ENGLISH

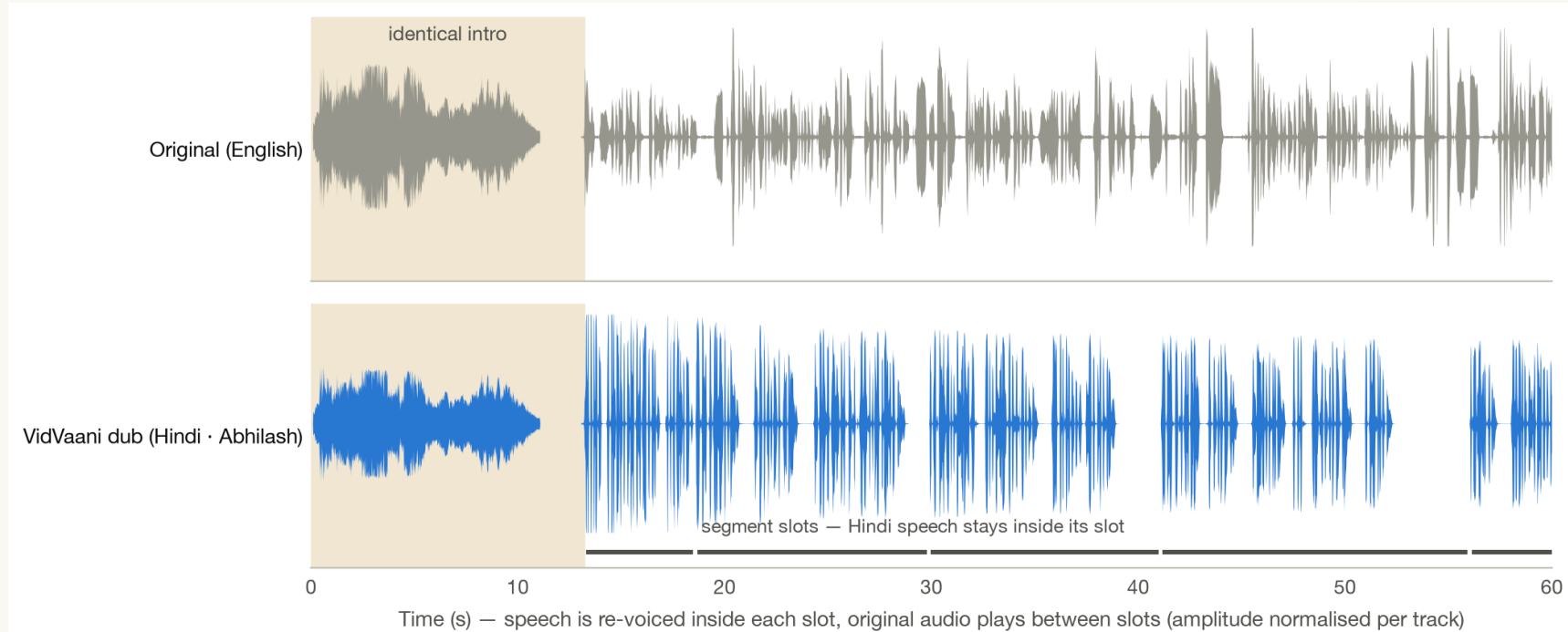
The x range is between zero to two and the y range is between zero to two.

HINDI

x और y की रेंज शून्य से दो तक है।

Terms such as **GeoGebra, x, y and PDF** are retained.
The explanation is translated into natural spoken Hindi.

Timing and assembly



Each translation receives a word budget. After synthesis, the clip is trimmed and its pace is adjusted; original audio remains in the gaps.

Demo runs stop at the requested segment

5

requested segments

58.5 s

final duration, with no English remainder

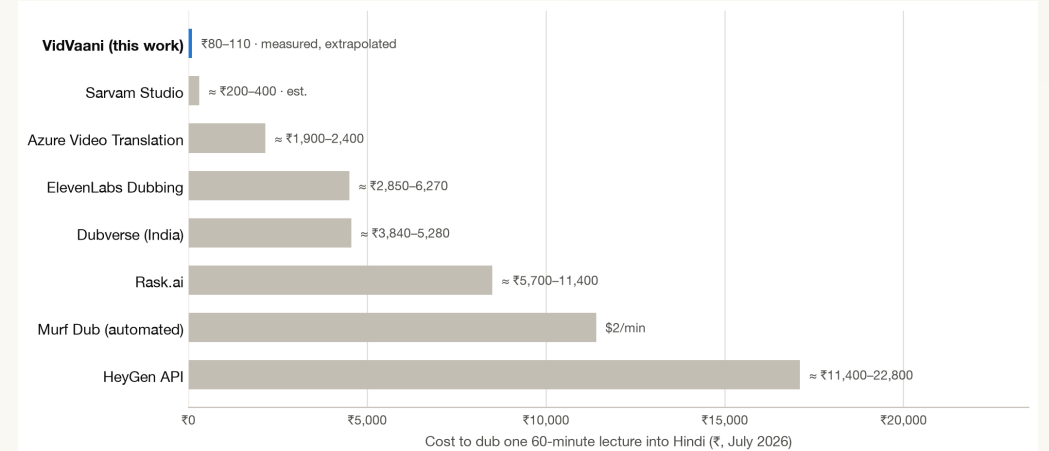
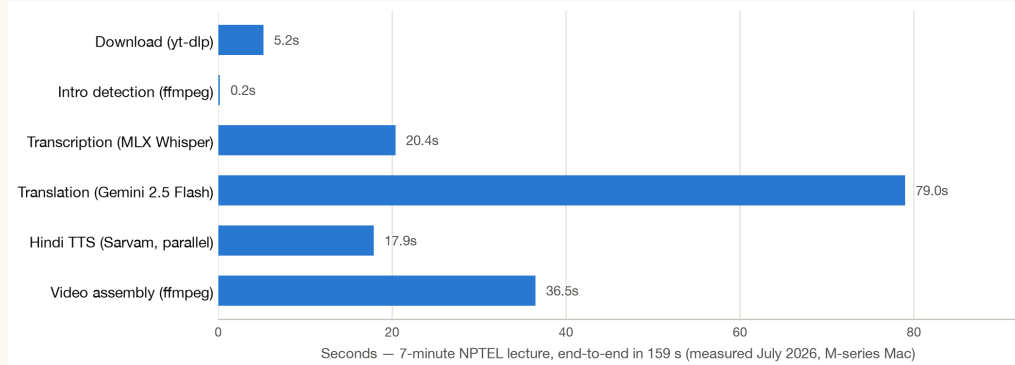
1864×1080

source resolution preserved

Demo mode ends after the last generated Hindi clip.
Full mode processes the complete source. The original video stream is copied without reducing its resolution.

```
uv run --frozen vidvaani-web  
# "Explore completed sample" makes no API calls
```


Measured time and cost



2½–4 min

end-to-end for a seven-minute lecture

₹80–110

extrapolated per lecture-hour

₹0

for a cached no-API demo run

Cloud and local speech options

CLOUD

Sarvam

Clear Hindi with Indian prosody and several voice choices. This is the default demonstration backend.

CLOUD

Gemini TTS

Natural delivery and style control, although its Hindi accent may sound less local.

LOCAL · ₹0

Gemma 4 + Qwen3-TTS

Translation and speech run on-device. Lecturer voice cloning is used only with consent.

59 s

Gemma 4 31B translation on the 58 s clip

1.6× RTF

Qwen3-TTS cloned speech generation

0.89

best consented speaker similarity; STT-checked

Demonstrations

[NIPUNBATRA.GITHUB.IO/VIDVAANI](https://nipunbatra.github.io/vidvaani)

Available comparisons

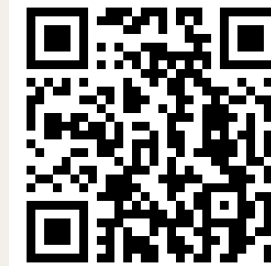
- Original English beside the Hindi result
- Sarvam, Gemini, fully local, and cloned-voice paths
- Short probability demo plus a full NPTEL lecture
- Methods, costs, privacy and limitations documented alongside the examples

The website supports audio comparison; the control room shows how each output was produced.

Acknowledgements

- Most of this material is based on the article "Deep Learning in Neural Networks: An Overview" by J. Schmidhuber^[1]
- The errors, if any, are due to me and I apologize for them
- Feel free to contact me if you think certain portions need to be corrected (please provide appropriate references)

कुछ आभार और अस्वीकरण। मैंने ज्यादातर सामग्री पहले पेपर से ली है। यह अठारह सौ इकहत्तर का है, इसलिए कुछ त्रुटियाँ हो सकती हैं। शायद मैंने कुछ तथ्य गलत बताए।



Limitations and next steps

QUALITY

Faculty review

The system produces a draft; subject experts must still review its pedagogical accuracy.

ALIGNMENT

Dense passages

Some segments require a shorter translation rather than faster synthesized speech.

TRUST

Voice consent

Voice cloning is opt-in, and speaker similarity alone is not sufficient for release.

Next steps are faculty-in-the-loop editing, a terminology glossary, native TTS pace control, support for more Indian languages, and evaluation of student comprehension.

From an English lecture to a Hindi version.

The pipeline is open, its intermediate results can be inspected, and the translation and speech stages can run locally when required.

Demo

nipunbatra.github.io/vidvaani

Code

github.com/nipunbatra/vidvaani

Question

Which lecture should be evaluated next?